Graphical Perception

Graphical Perception

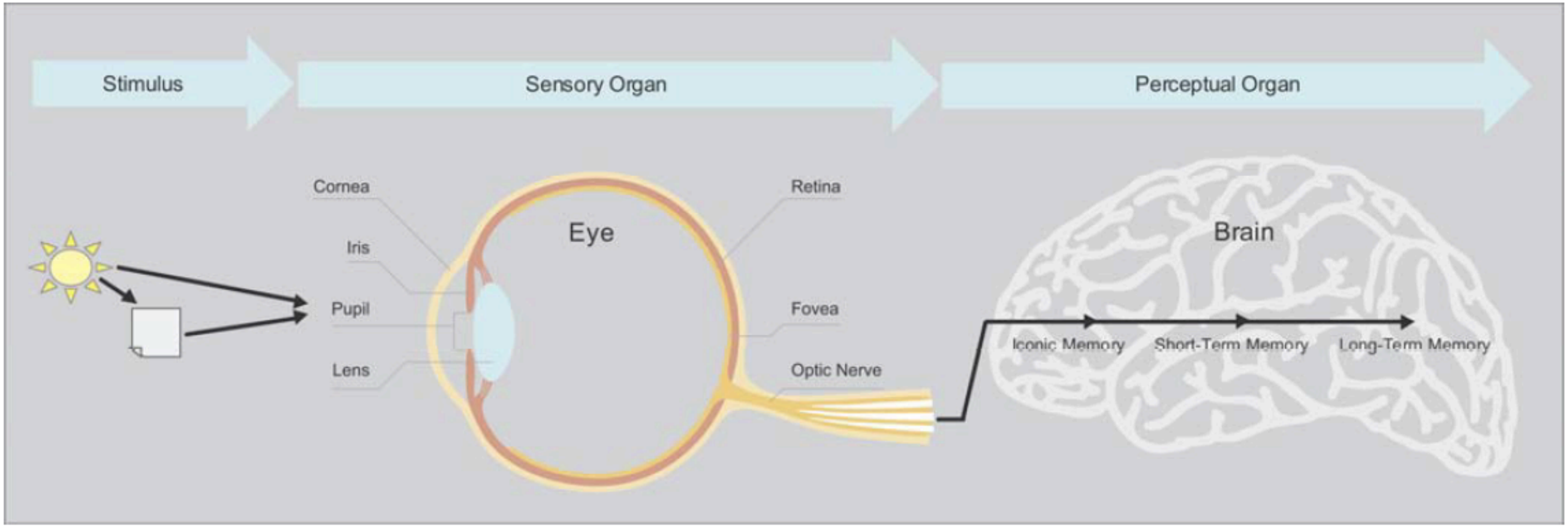
"If we can understand how perception works, our knowledge can be translated into rules for displaying information.

Following perception-based rules, we can present our data in such a way that the important and informative patterns stand out.

If we disobey the rules, our data will be incrompehensible or misleading."

(C. Ware 2004: xxi)

How Perception Works

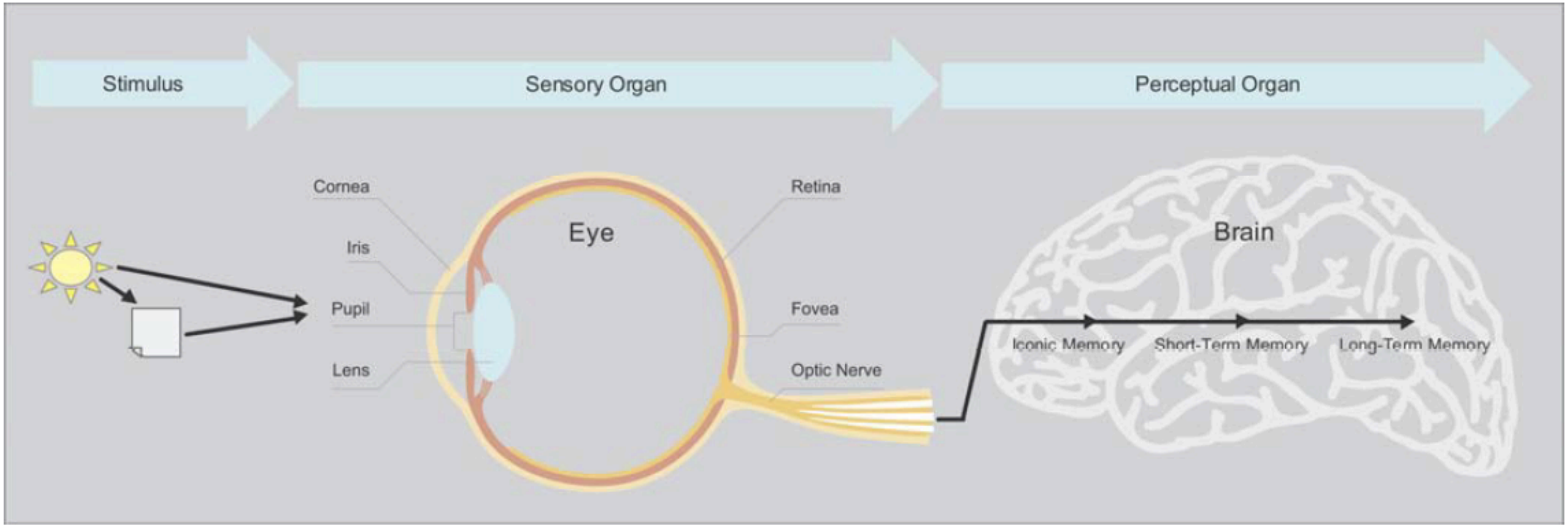


Light passes through the pupil and hits the retina – a thin surface coated with millions of nerves called rods and cones.

Rods and cones translate what they detect into neural signals and pass them on to the brain.

We don't see things with our eyes – we see them with our brains!

How Perception Works



Iconic Memory (Visual Sensory Register): Pre-attentive processing (unconscious and automatic) of visual attributes (colors, shapes, etc.)

Short Term or Working Memory: Attentive processing of "chunks" of information

Long Term Memory: Storing and recognizing familiar patterns

Attributes of Pre-attentive Processing

756395068473

658663037576

860372658602

846589107830

How many 3s are there?

Attributes of Pre-attentive Processing

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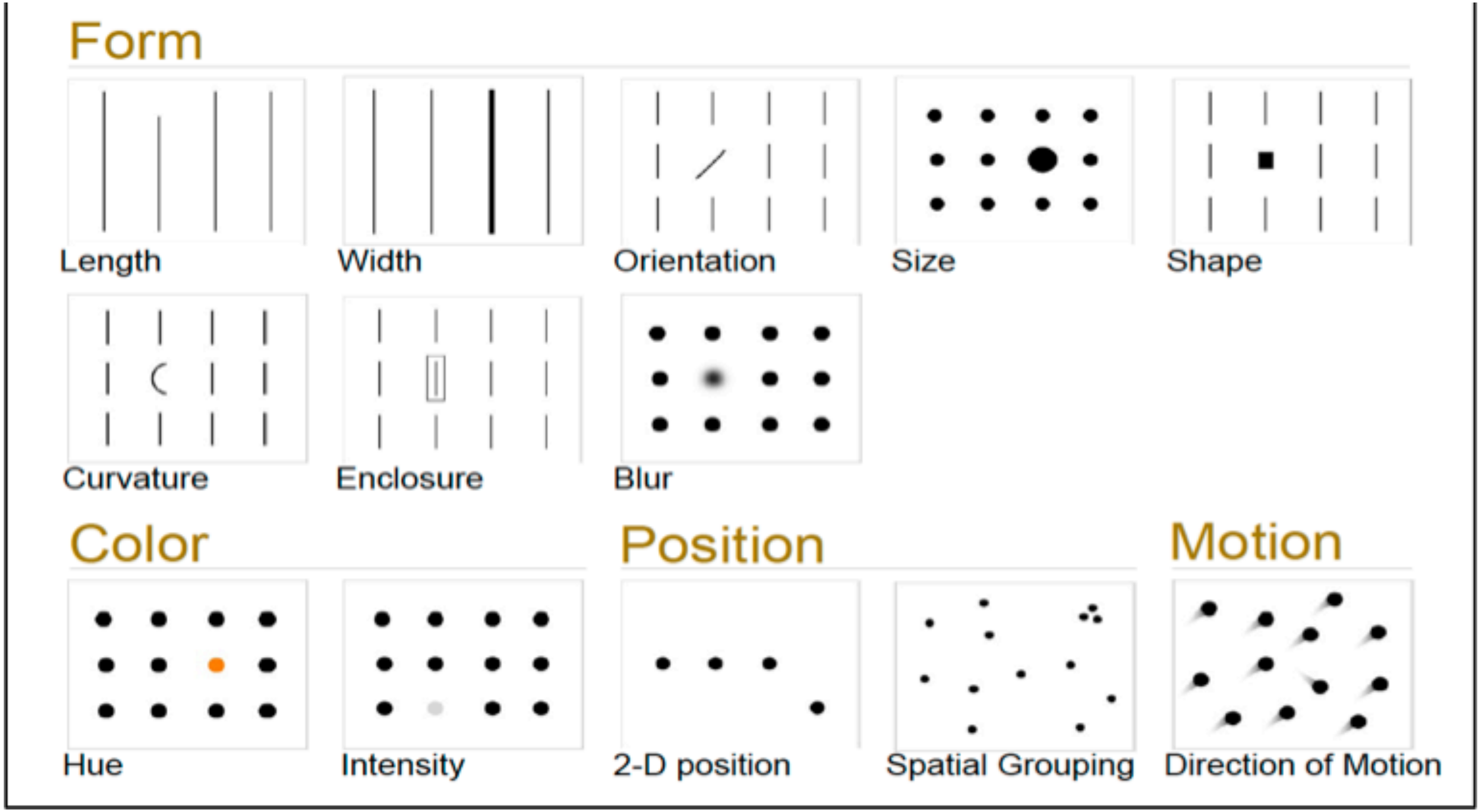


Attributes of Pre-attentive Processing

We can use attributes of pre-attentive processing to

1. Choose visual encodings of abstract data (choice of graphical formats)

2. Direct audience attention (design choice)

3. Create a visual hierarchy (design choice)

Attributes of Pre-attentive Processing

Attributes of Pre-attentive Processing

Pre-attentive attributes become less distinct as the variety of distractors increases.

How much bigger is the second circle?

Attributes of Pre-attentive Processing

Bring the colors in an order from small to large!

Attributes of Pre-attentive Processing

Visual attributes are not all created equal…

Attribute	*Quantitatively Perceived?*
2-D Position	Yes
Length	Yes
Width	Limited
Size	Limited
Color Intensity	Limited
Orientation	No
Shape	No
Enclosure	No
Color Hue	No

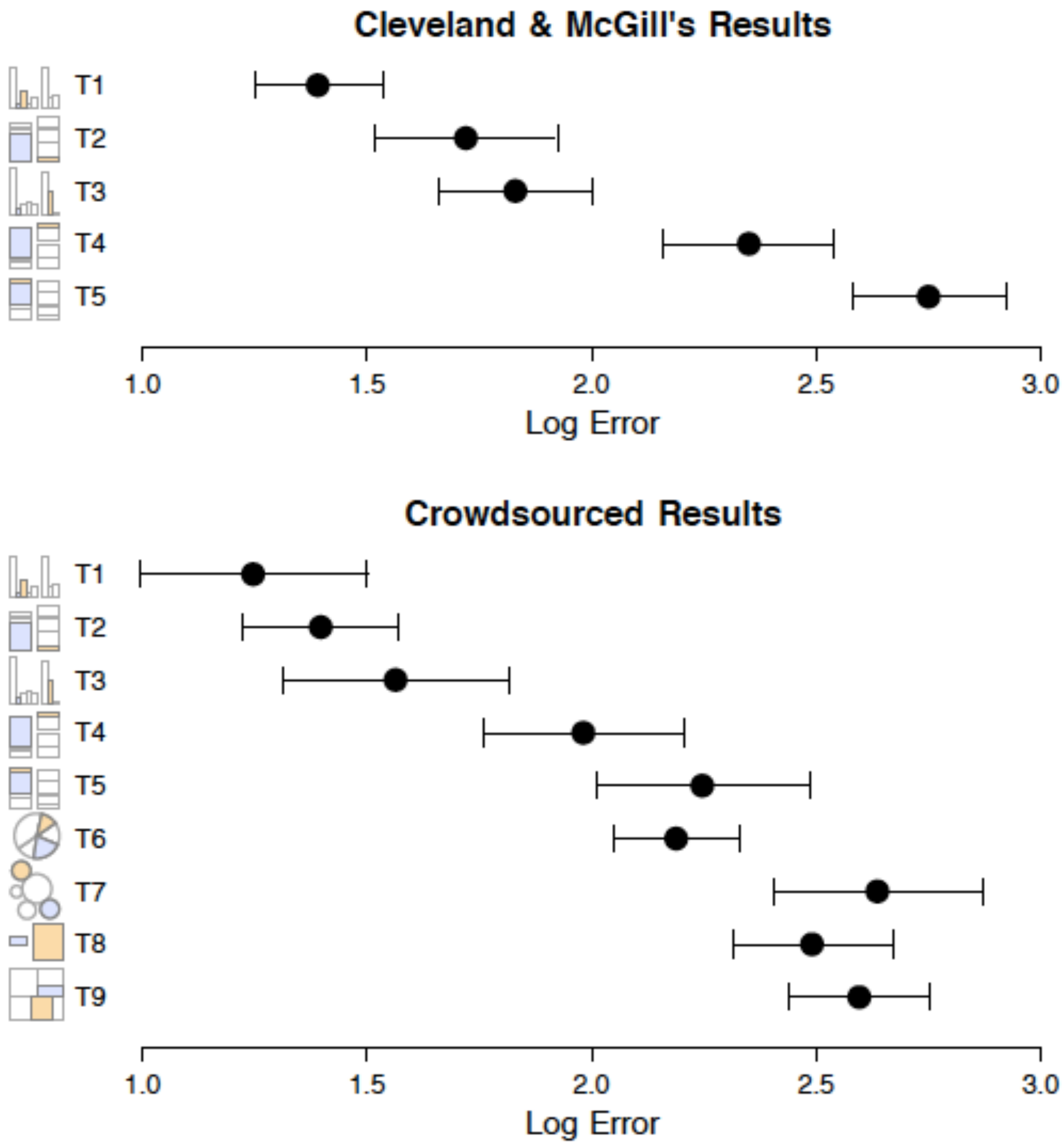
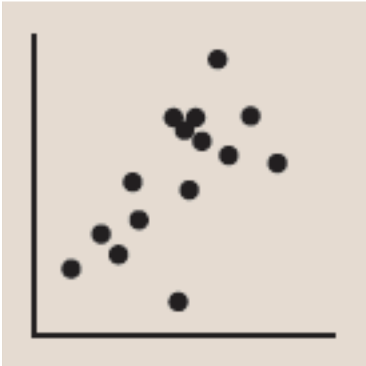
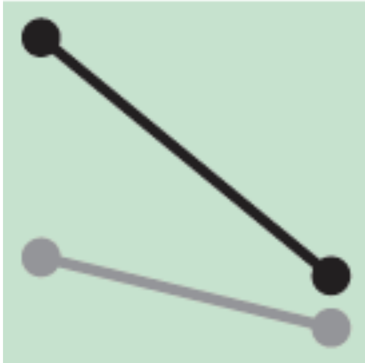


Figure 4: Proportional judgment results (Exp. 1A & B). Top: Cleveland & McGill's [7] lab study. Bottom: MTurk studies. Error bars indicate 95% confidence intervals.

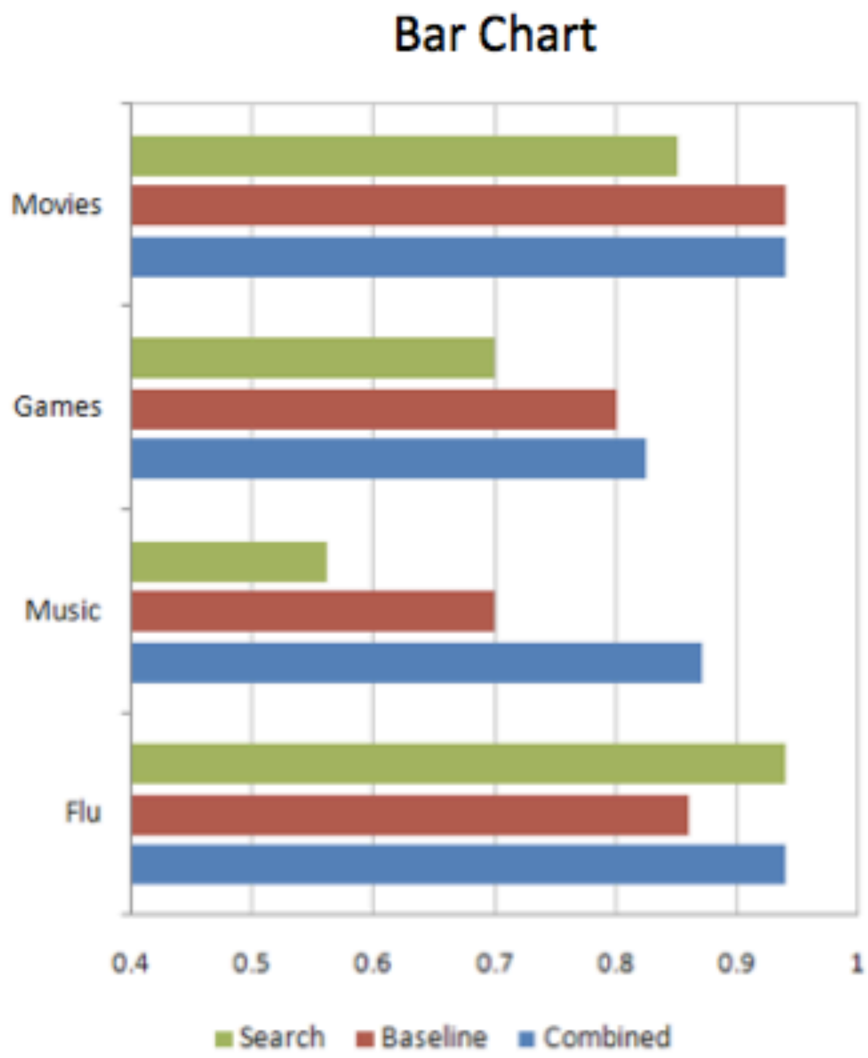
(Heer & Bostock 2010)

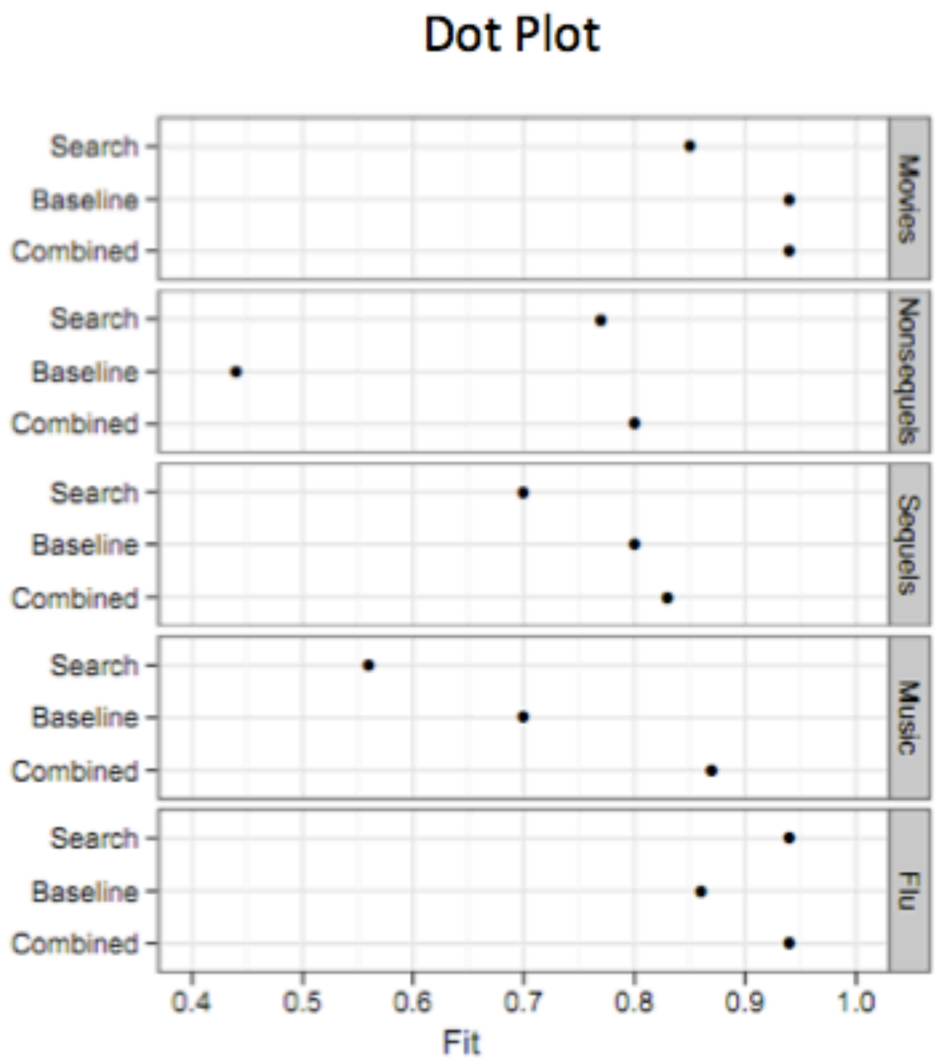
Now you know why…

Dot Chart/Scatter Plot:	2-D position of visual objects
Line Chart:	2-D position, connected to give shape to a series of values
Bar Chart:	Length and 2-D position

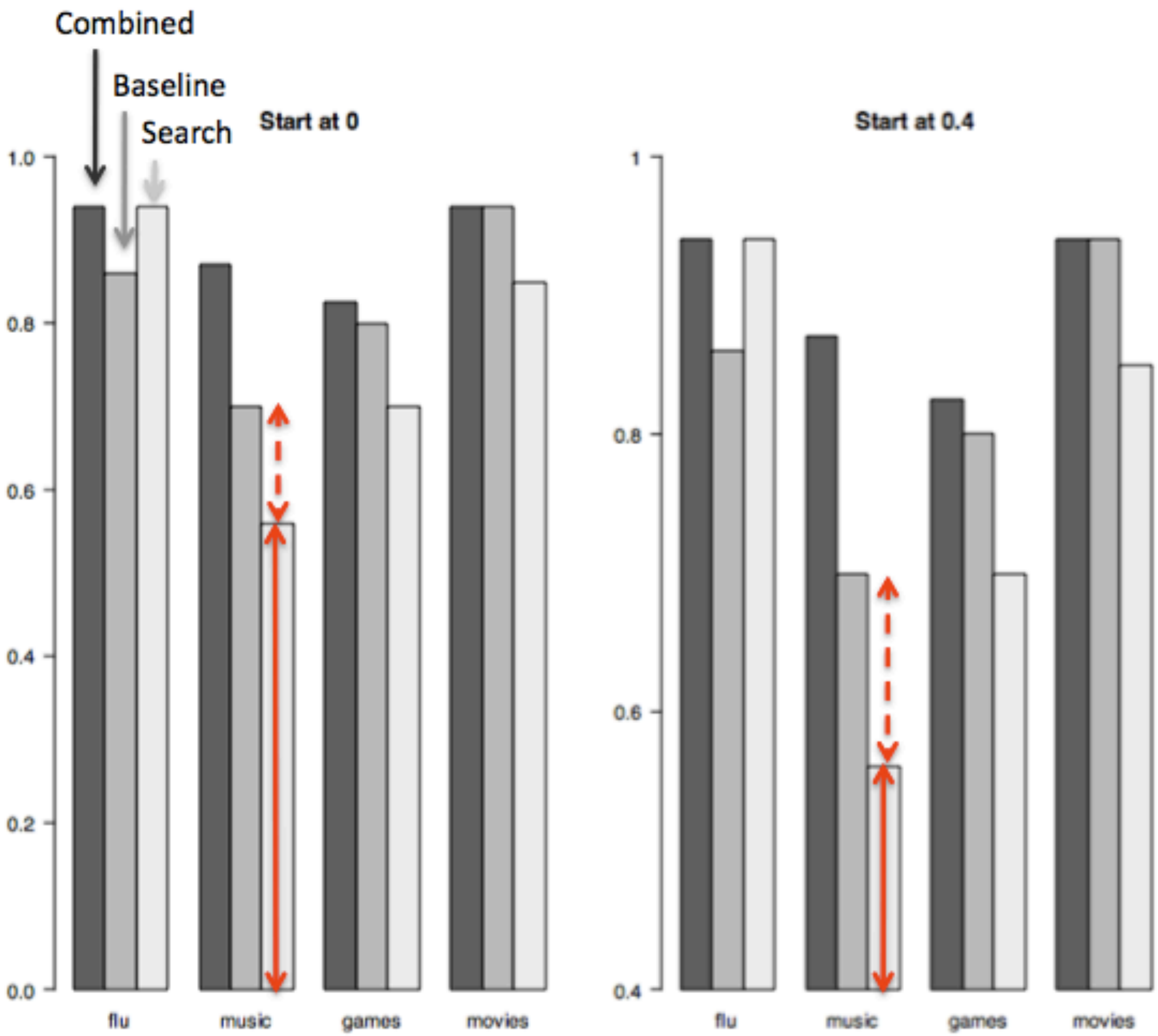
Trade-Offs and Comparative Advantages



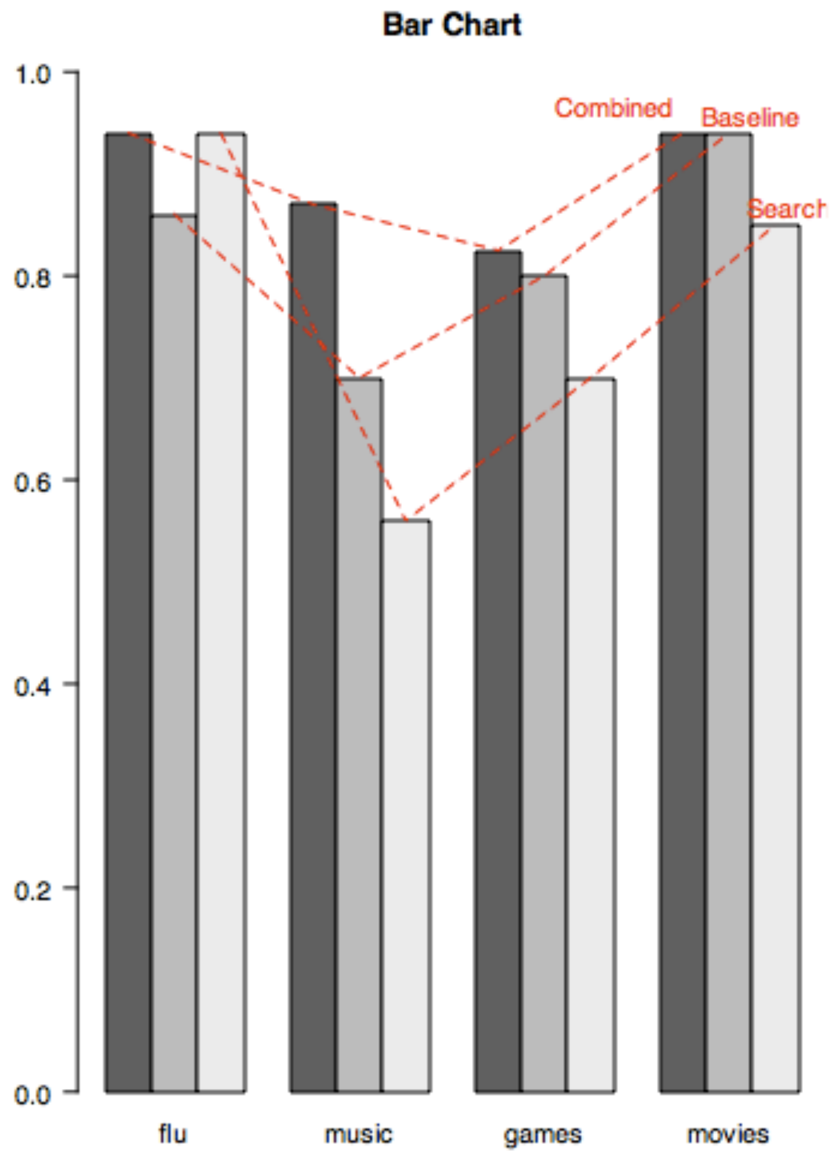


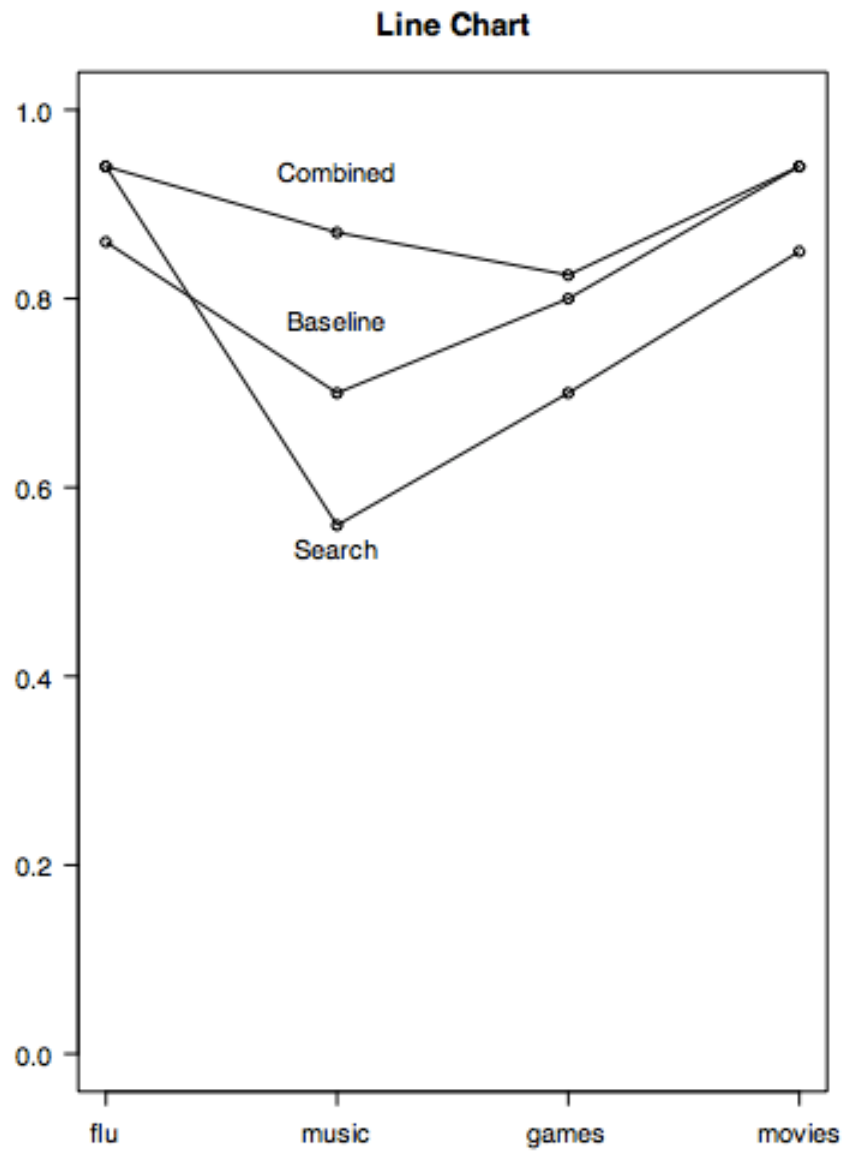
Source: Junkcharts.

Trade-Offs and Comparative Advantages

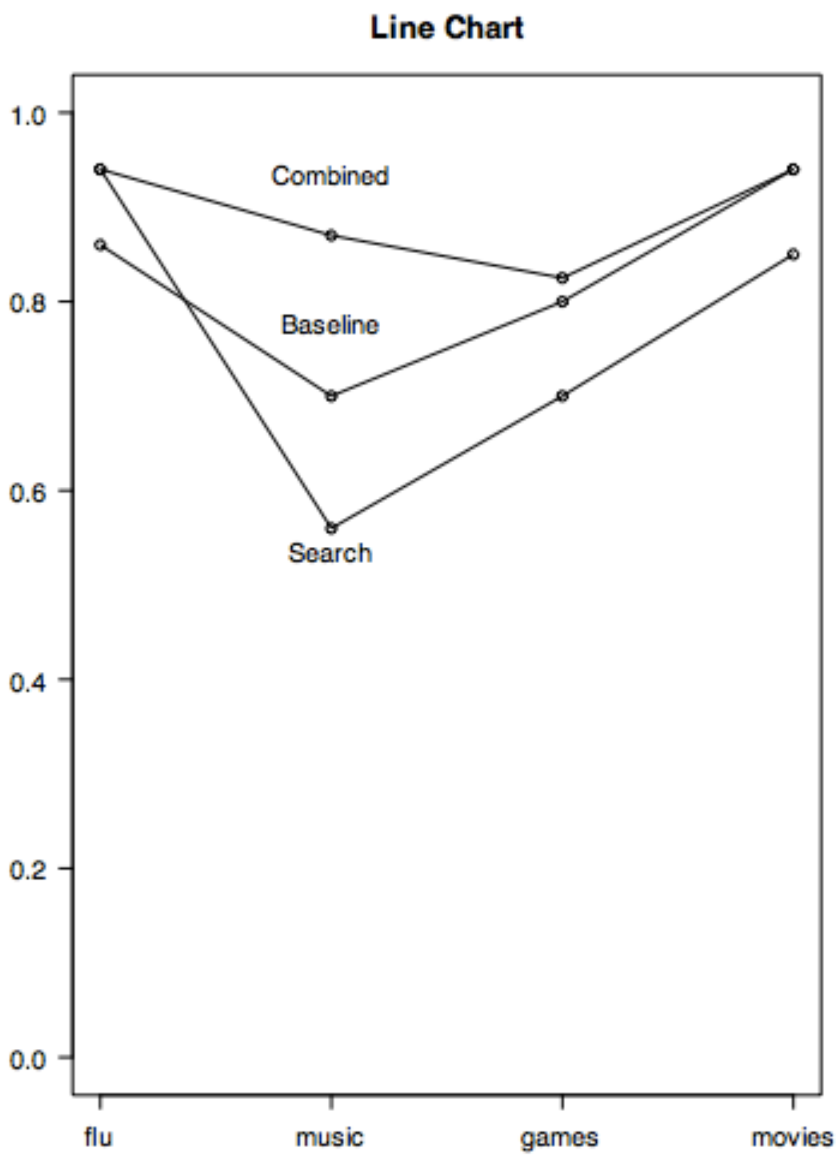


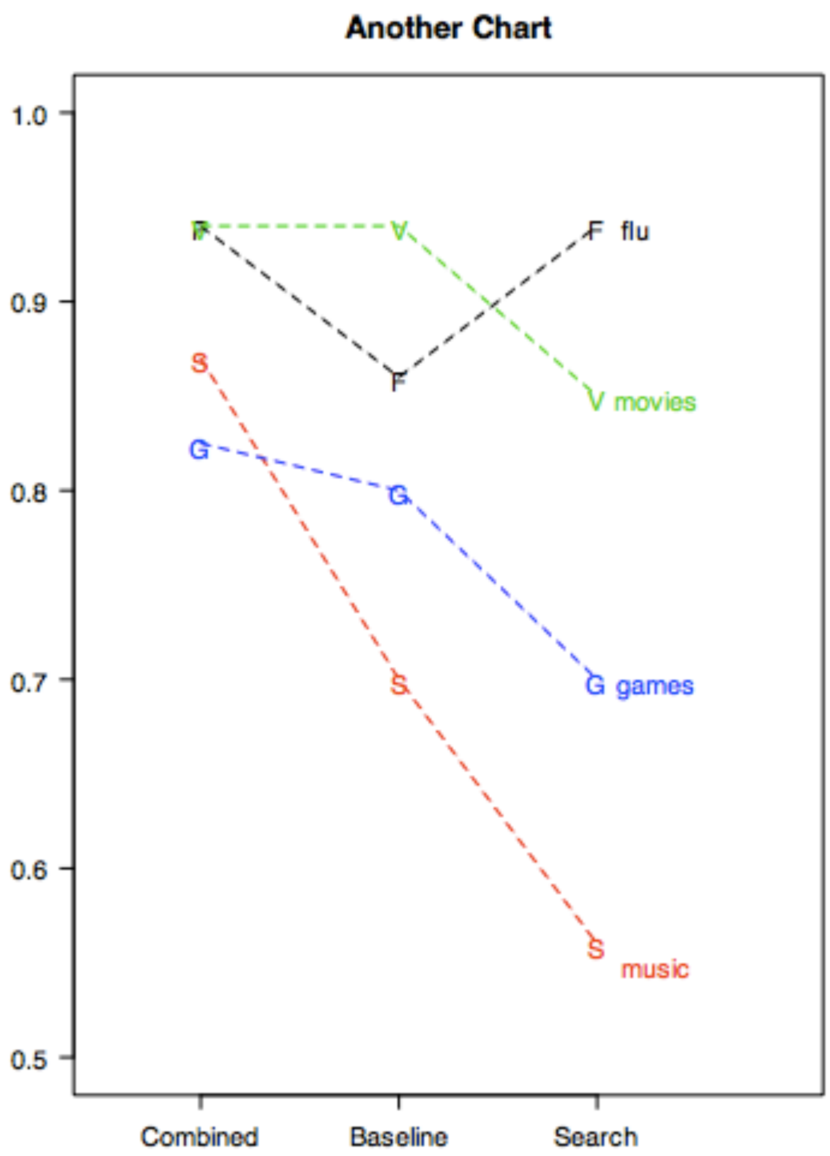
Trade-Offs and Comparative Advantages





Trade-Offs and Comparative Advantages





Gestalt Principles of Visual Perception

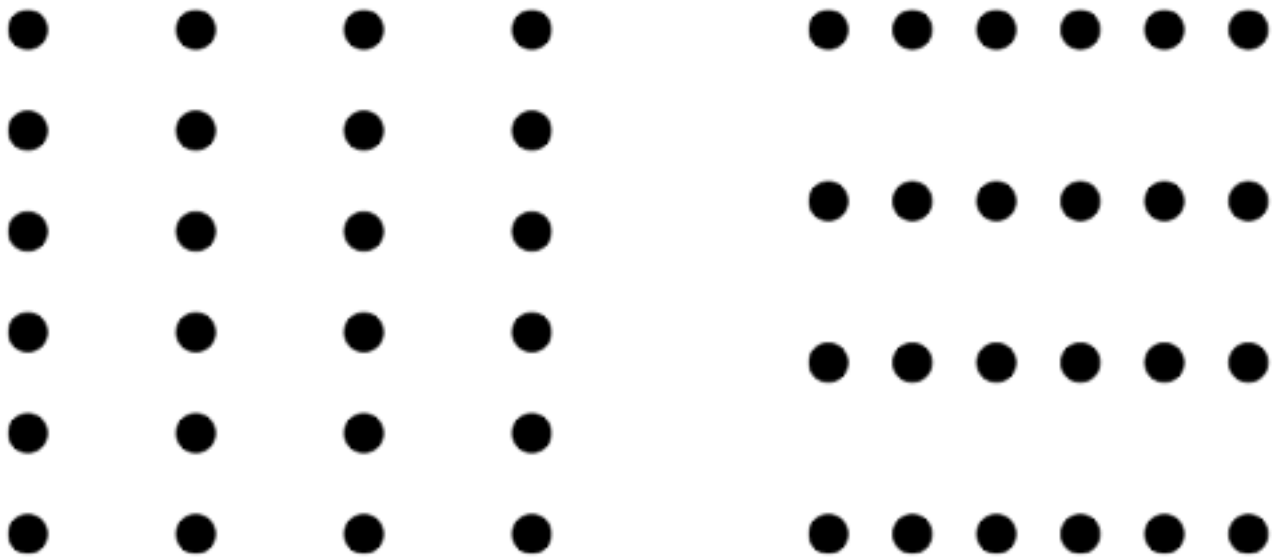
Visualization works because it abstracts from single data points and leverages the human brain´s ability to form these data points into emergent patterns.

This ability has been studied by the Gestalt School of Psychology in the early 1900s.

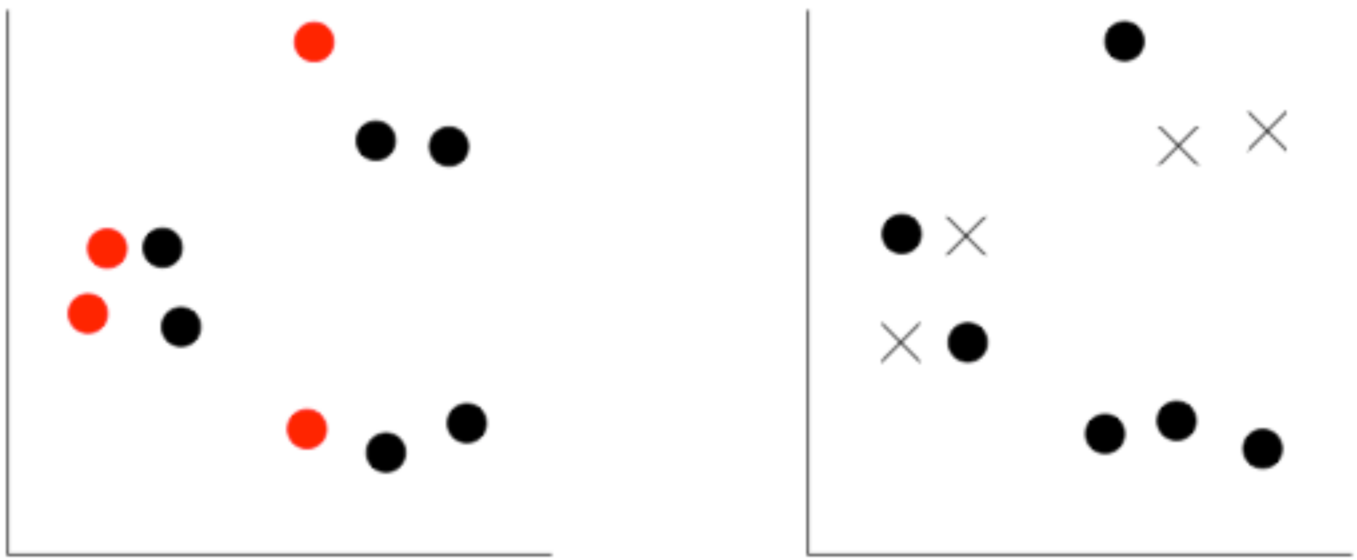
Understanding this ability allows us to use it strategically in the design of visualizations.

Gestalt Principles of Visual Perception

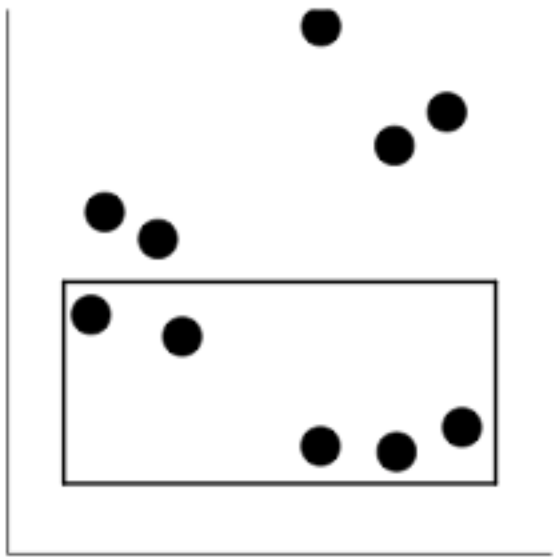
Proximity: Objects that are close together are perceived as a group.

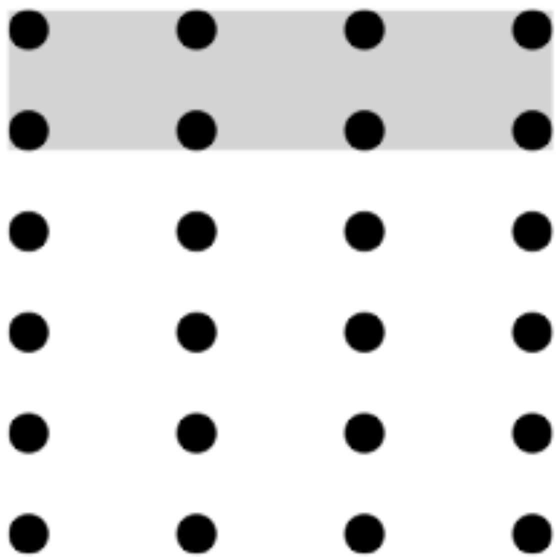


Similarity: Objects that share similar attributes (e.g. color or shape) are perceived as a group.



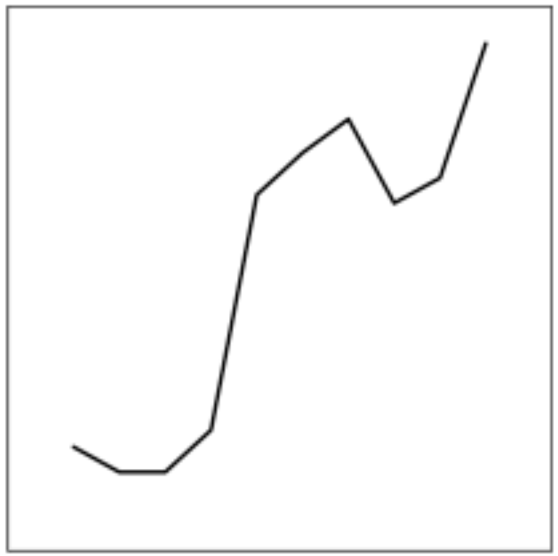
Enclosure: Objects that have boundary around them are perceived as a group.

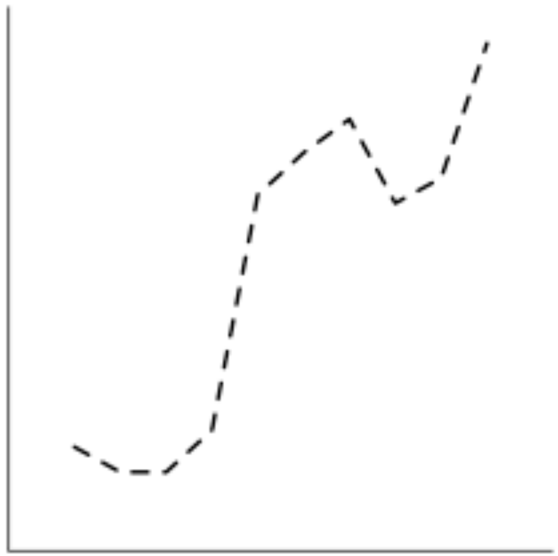




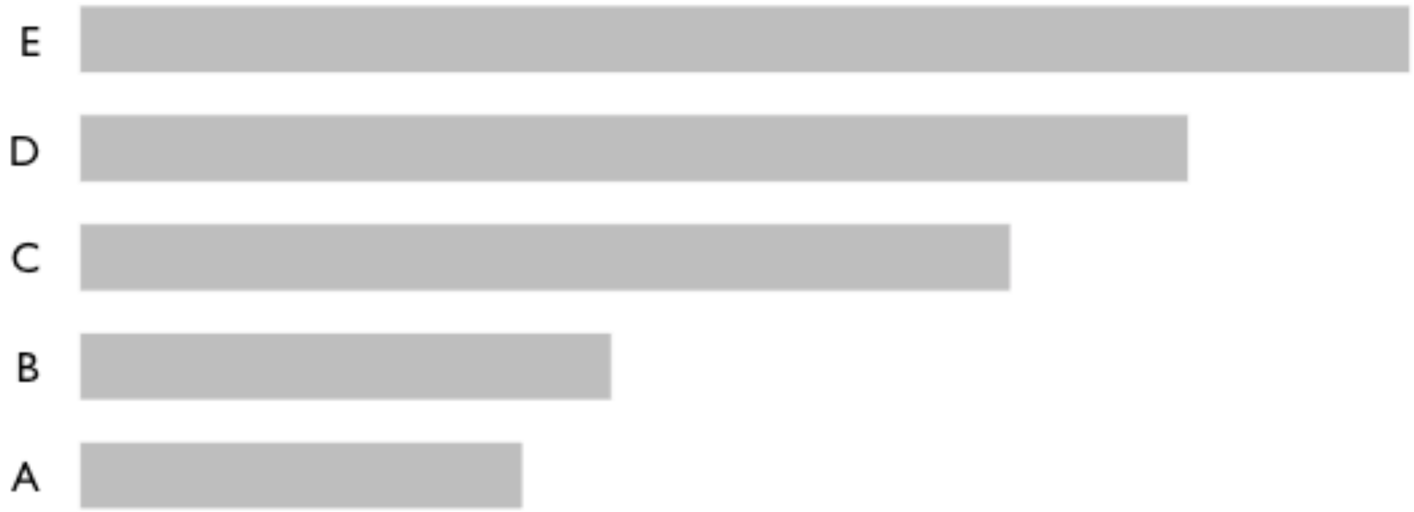
Gestalt Principles of Visual Perception

Closure: Open structures are perceived as closed, complete, and regular.

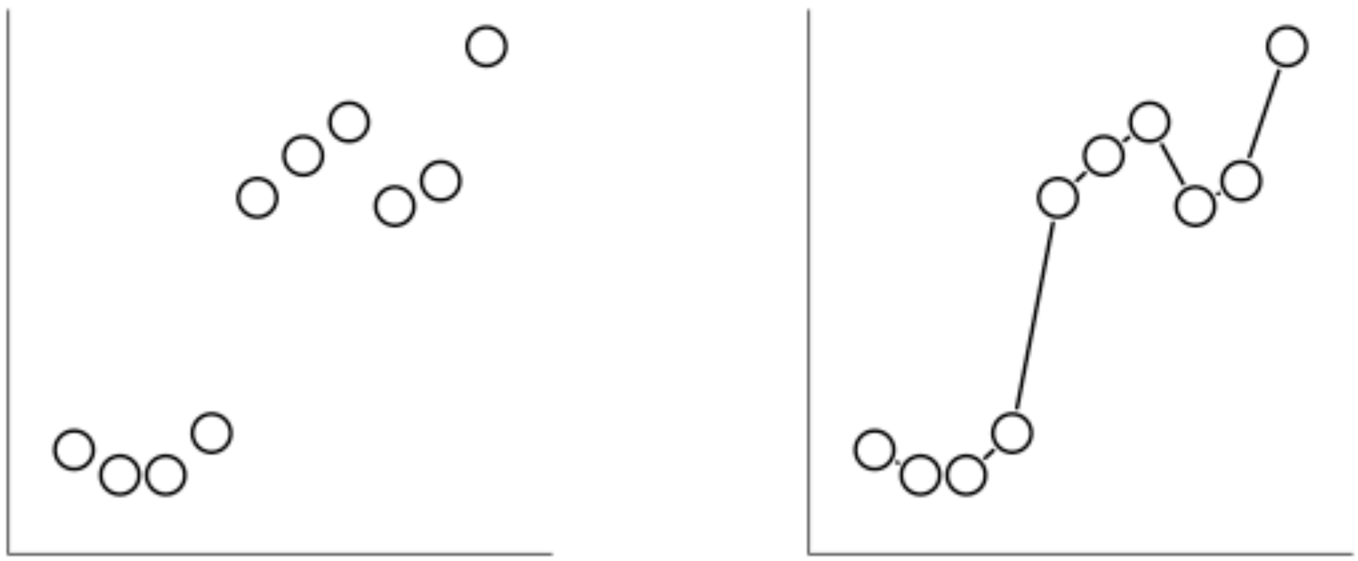




Continuity: Objects that are aligned together are perceived as a group.



Connection: Objects that are connected (e.g. by a line) are perceived as a group.



The Zen of Visualization Design

Above else show the data.

Maximize the data-ink ratio.

Erase non-data-ink.

Erase redundant data-ink.

Revise and edit.

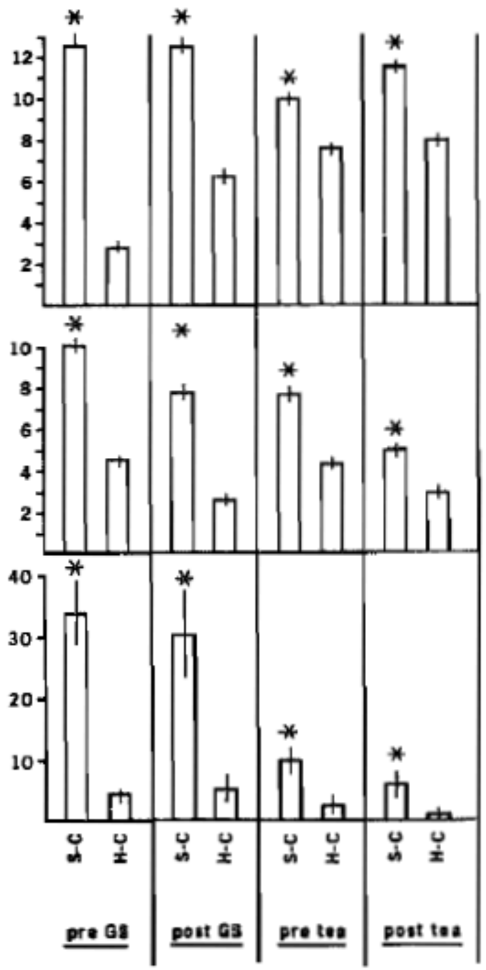
The Zen of Visualization Design

$$\text{Data-ink Ratio} = \frac{\text{data ink}}{\text{total ink used to print the graphic}}$$

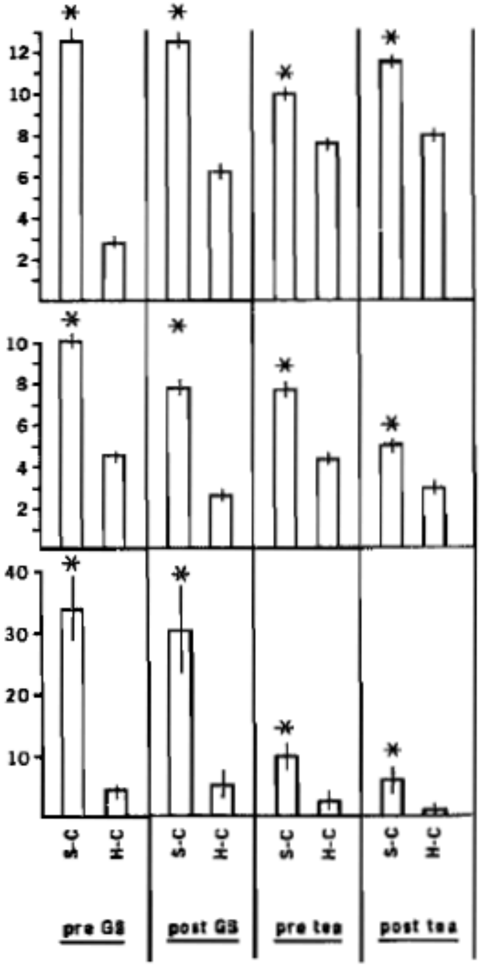
= proportion of ink devoted to the non-redundant display of data-information

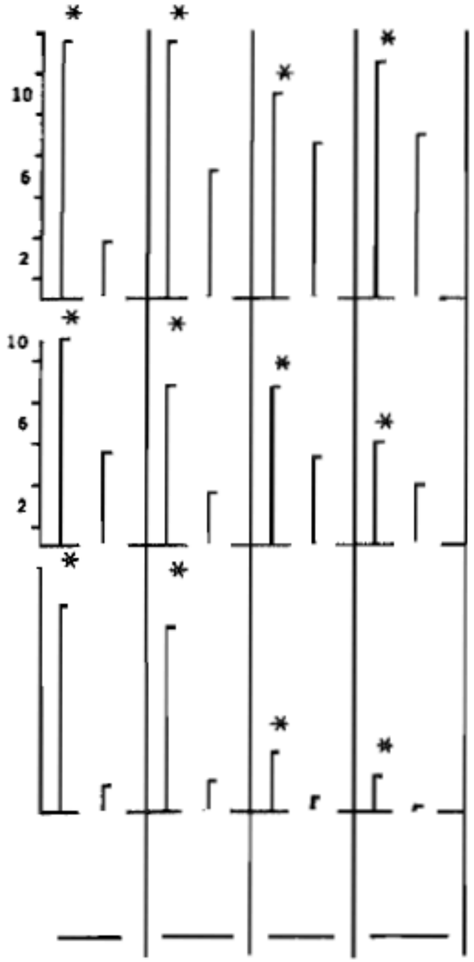
= 1 – proportion of a graphic that can be erased without loss of data-information

(Radical) Tufte Example

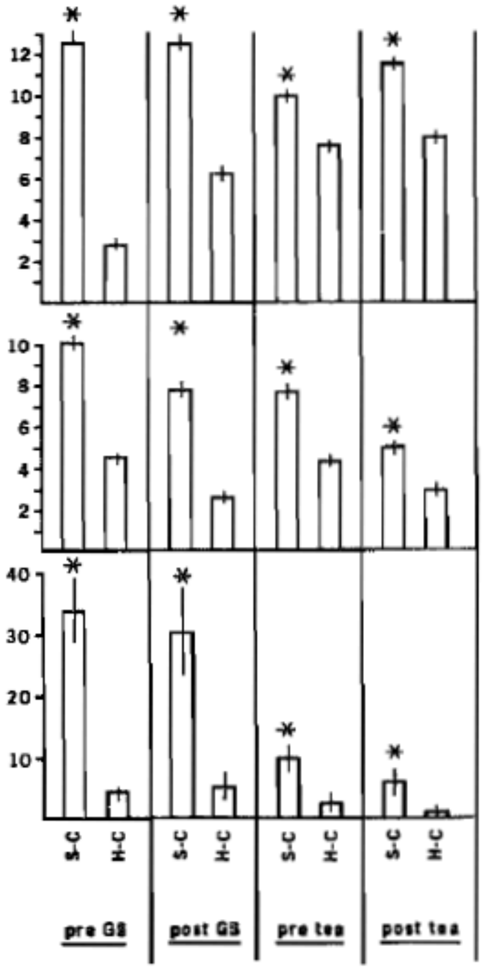


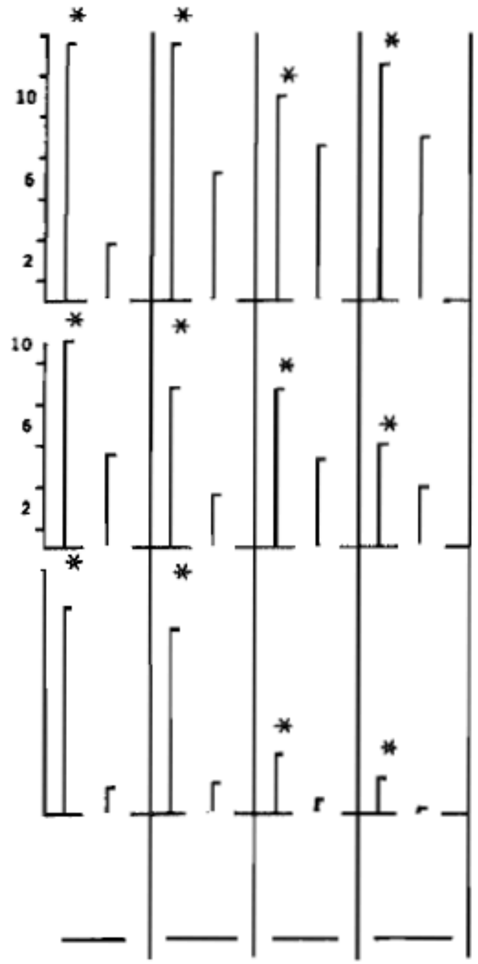
(Radical) Tufte Example

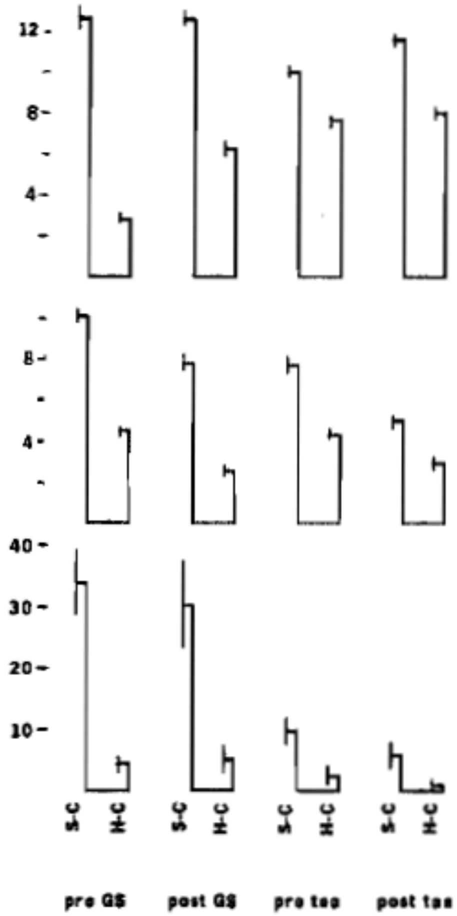




(Radical) Tufte Example







Stephen Few's Design Recommendation

Reduce the Non-Data Pixels

1. Subtract unnecessary non-data pixels.

   Ask yourself: "Would the data suffer any loss of meaning or impact if this were eliminated?"

   If the answer is "no," then get rid of it.

2. De-emphasize and regularize the remaining non-data pixels.

   e.g. use thin lines and light grey for supporting non-data components of the graph (axes, labels, etc.)

Stephen Few's Design Recommendation

Enhance the Data Pixels

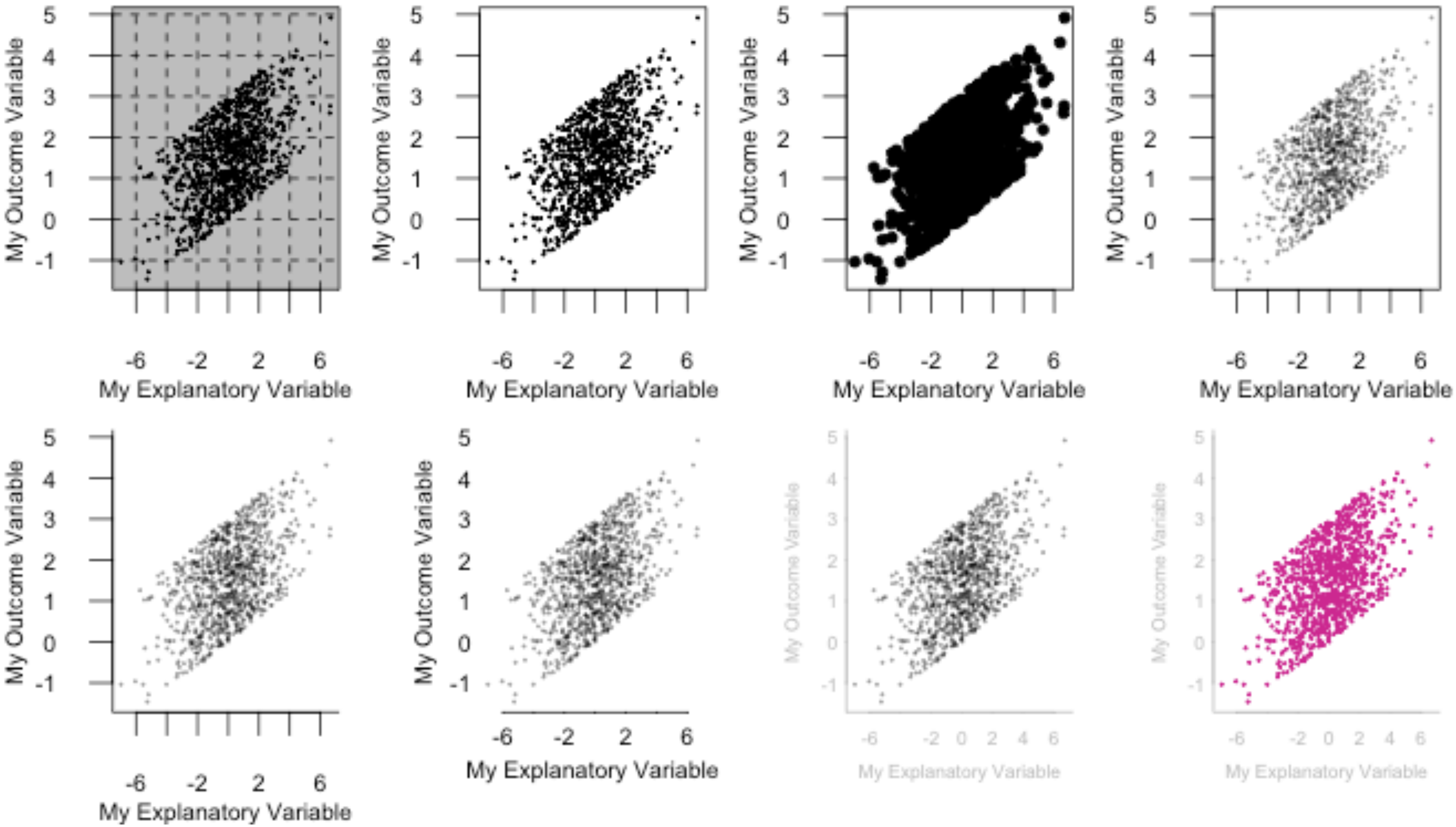
1. Subtract unnecessary data pixels

Not all information is equally important.

2. Emphasize the most important data pixels

Use attributes of pre-attentive processing (e.g. color, size, width) to emphasize the most important data pixels.

Example


My Outcome Variable
5 4 3 2 1 0 -1
-6 -2 2 6
My Explanatory Variable
My Outcome Variable
5 4 3 2 1 0 -1
-6 -2 2 6
My Explanatory Variable
My Outcome Variable
5 4 3 2 1 0 -1
-6 -2 2 6
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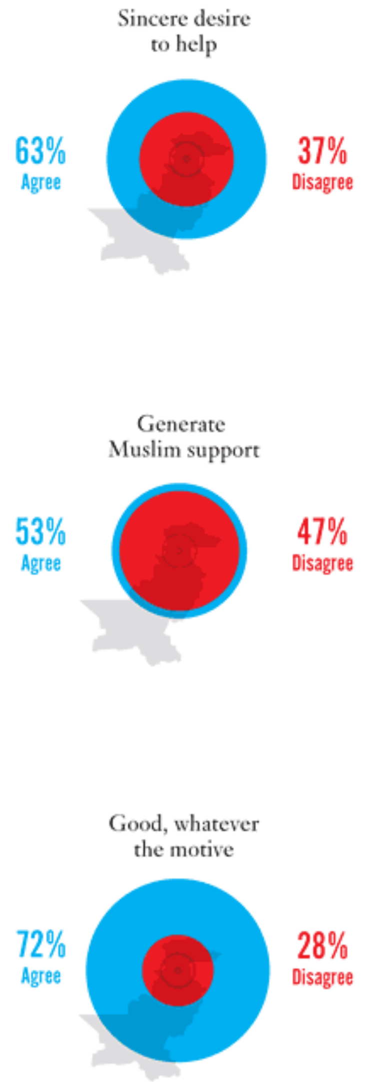
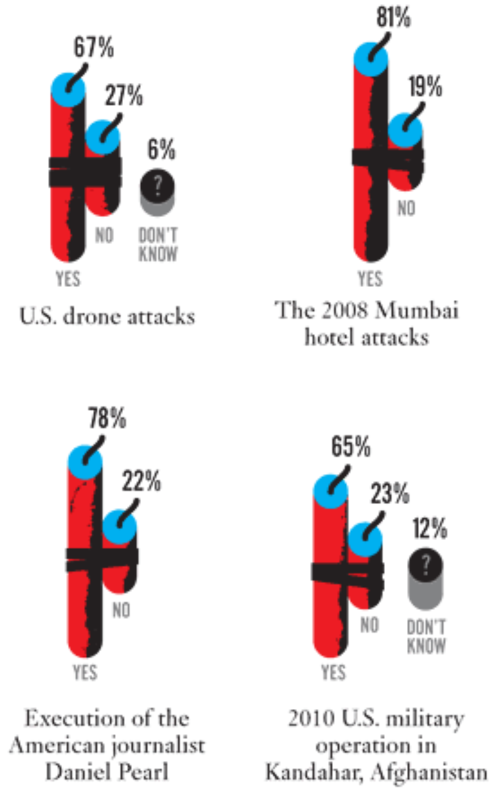
My Explanatory Variable
My Outcome Variable
5 4 3 2 1 0 -1
-6 -2 0 2 4 6
My Explanatory Variable

Earthquake Aid

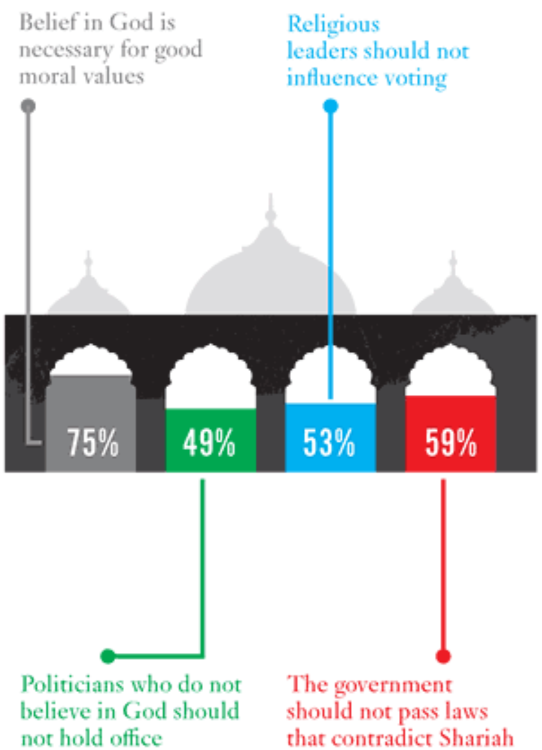
WHY DID THE U.S. GIVE MILLIONS
AFTER THE 2005 KASHMIR EARTHQUAKE?

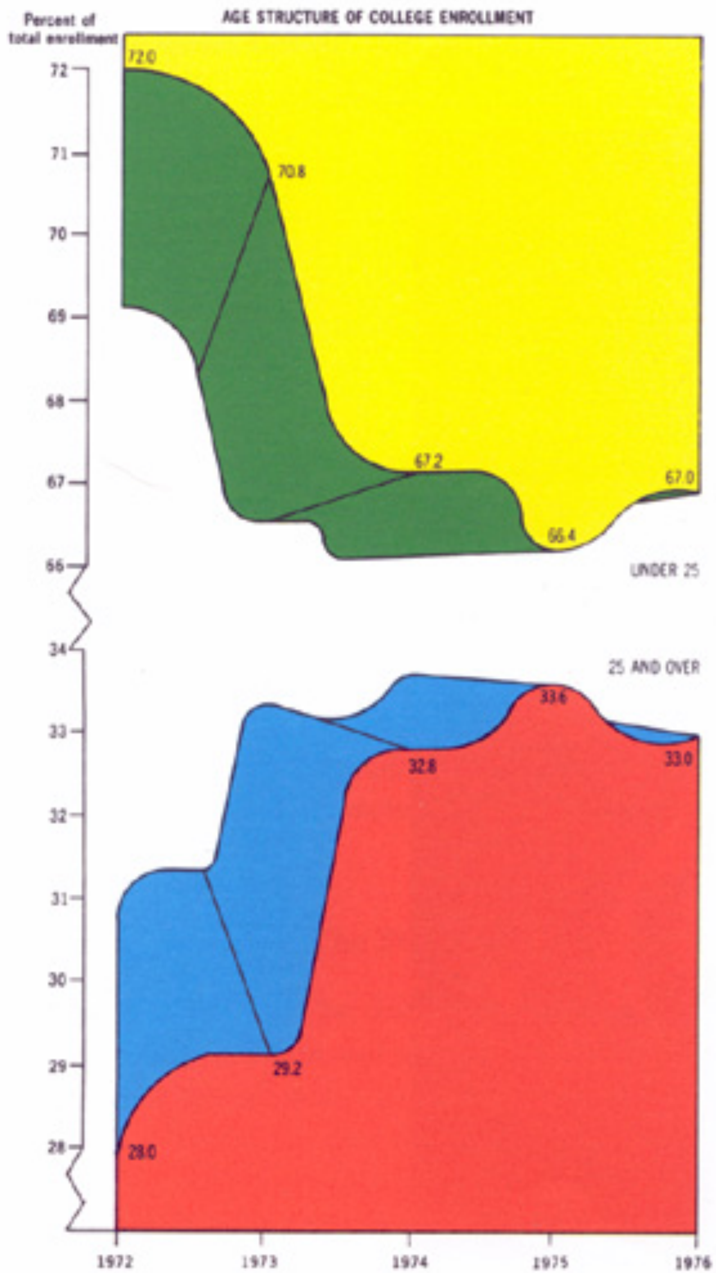


Which Is an Act of Terrorism?



Religion




AGE STRUCTURE OF COLLEGE ENROLLMENT
Percent of total enrollment
72.0
70.8
67.2
66.4
67.0
UNDER 25
25 AND OVER
33.6
32.8
33.0
29.2
28.0
1972
1973
1974
1975
1976